

Patient-level coronary artery disease classification from cardiac MRI using out-of-fold heart segmentation and same-slice region controls

A proof-of-concept study of a 30-patient public data-set – JPEG release

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Scope: This manuscript reports a secondary, proof-of-concept analysis of released images. It does not present a clinically validated diagnostic system and should not be used for patient care.

Result provenance: All numerical results in this manuscript are from the complete 5 October 2026 execution with 5,097 accepted targets. Changed targets forced all five segmentation folds and all out-of-fold parts to be recomputed; a changed semantic feature fingerprint then forced all eleven CPU/float32 feature representations to be rebuilt. Earlier version 1.1 values are superseded and are not combined with the current run.

This version has not undergone peer review.

Abstract

Background: Medical-image classifiers can obtain high apparent performance from correlated slices, acquisition patterns, or anatomy outside the intended target. The downloaded CAD Cardiac MRI Dataset release contained 63,425 JPEG images but only 30 independent top-level patient folders. This study evaluated whether cardiac-region representations were more informative than exact same-slice full-image and extra-cardiac controls under patient-level validation.

Methods: Each Directory_* folder was treated as one patient (16 Normal, 14 Sick), as confirmed by the dataset creator. The current analysis used five patient-level cross-fitted 2.5D Attention U-Nets trained from 5,097 accepted targets (3,949 heart-present and 1,148 no-heart). Every released image received an out-of-fold mask from a model that did not train on its patient. A frozen ImageNet-pretrained EfficientNet-B0 encoded full-image, cardiac-ROI, and exact-complement representations. Slice embeddings were averaged first within folder-defined series and then across series for each patient. Nested patient-level cross-validation evaluated eleven representations using exact same-slice controls, a secondary cross-class matched cohort, and paired patient-bootstrap intervals.

Results: The expanded manual-target state caused all five segmentation folds and all five out-of-fold prediction parts to be recomputed. Out-of-fold prediction produced 63,107 final-valid masks among 63,425 images; positive-target patient-balanced Dice was 0.809, and 73.4% of accepted no-heart targets received empty masks. On exactly 50,735 attention-eligible slices, the automatic cardiac ROI achieved AUC 0.938 (95% CI 0.847-1.000), compared with 0.830 for the full image and 0.772 for the complement. Paired AUC differences were 0.107 (95% CI 0.009-0.235) versus the full image and 0.165 (0.029-0.344) versus the complement. In the matched cohort, ROI AUC was 0.942, versus 0.821 for the full image and 0.741 for the complement. On the manual-positive subset, automatic and manual ROIs achieved AUCs of 0.973 and 0.951, respectively; their paired difference did not exclude zero.

Conclusions: After expanded manual annotation and a complete rerun of every dependent stage, out-of-fold cardiac localization retained an advantage over exact same-slice and matched controls. Complement AUCs of 0.741-0.835 nevertheless show substantial residual predictive information outside the intended cardiac region. With only 30 patients, no external validation, JPEG rather than DICOM data, and outcome-informed matching in a secondary analysis, these findings support a methodological proof of concept rather than clinical use.

Keywords: coronary artery disease; cardiac magnetic resonance; medical imaging; Attention U-Net; EfficientNet; patient-level cross-validation; out-of-fold segmentation; data leakage; shortcut learning; reproducibility

Key findings

- The effective independent sample size was 30 patients, not 63,425 images; every image from one patient was kept within one fold.
- On exactly the same 50,735 slices, the automatic cardiac ROI outperformed both the full image and its complement at patient level.
- Complement AUCs from 0.741 to 0.835 indicate substantial predictive information outside the intended cardiac region and argue against clinical overinterpretation.
- On the same 3,949 annotated images, automatic and manual ROIs achieved AUCs of 0.973 and 0.951; the paired difference was -0.022 (95% CI -0.081 to 0.009) and did not establish equivalence.
- The result-associated 5 October pipeline used semantic fingerprints: changed targets retrained all five segmentation folds, recomputed all five out-of-fold parts, and rebuilt all eleven feature representations before evaluation.

Abbreviations

AUC, area under the receiver operating characteristic curve; AP, average precision; CAD, coronary artery disease; CI, confidence interval; CMR, cardiac magnetic resonance; CV, cross-validation; OOF, out-of-fold; PCA, principal component analysis; ROI, region of interest; TTA, test-time augmentation.

1. Introduction

Coronary artery disease (CAD) remains a major target for non-invasive imaging and automated decision support. Cardiac magnetic resonance (CMR) provides multiparametric information about anatomy, function, perfusion, scar, and tissue characteristics without ionizing radiation. Khozeimeh and colleagues released a CMR image dataset and reported a hybrid convolutional-neural-network/random-forest approach for CAD classification [1]. The associated public Kaggle resource is distributed as JPEG images in Normal and Sick folders [2].

A central difficulty in image-based machine learning is defining the independent unit of analysis. Thousands of neighbouring frames from one patient are strongly correlated; placing slices from the same patient in both training and testing can produce severe leakage and overly optimistic performance [9]. Small patient samples can also yield large uncertainty even when image counts are very high [11]. During the present project, a structural audit showed that the downloaded release contained 63,425 images but only 30 top-level Directory_* folders. The dataset creator subsequently confirmed that each Directory_* folder represents one patient and that the Kaggle release is a subset of the larger cohort described in the associated article (personal correspondence, 2 September 2026).

A second difficulty is shortcut learning: a classifier may use scanner style, JPEG export geometry, sequence composition, padding, mask shape, or extra-cardiac anatomy rather than disease-related cardiac information [10]. A high AUC alone therefore does not establish that a model has learned a clinically meaningful representation. The project was redesigned around patient-level folds, out-of-fold segmentation, exact same-slice controls, complement regions, and paired patient-level comparisons.

The primary objective was to test whether an automatically segmented cardiac ROI provides stronger patient-level discrimination than (i) the full image and (ii) the pixels outside the ROI, when all three representations are built from the same source slices. Secondary objectives were to evaluate matched Sick/Normal cohorts, compare automatic and researcher-created masks on the same images, quantify out-of-fold segmentation quality, and document residual evidence of dataset-specific shortcuts. The study follows the transparency principles of CLAIM and TRIPOD+AI where applicable to a proof-of-concept secondary analysis [12,13].

2. Methods

2.1 Study design and analysis hierarchy

This was a retrospective secondary analysis of a publicly released JPEG dataset. The released class label was Normal or Sick; Sick was interpreted as the CAD-labelled class in accordance with the associated publication [1]. The patient, not the slice, was the independent evaluation unit. The prespecified primary comparison comprised B2 (full image), AU1 (automatic cardiac ROI), and C1 (the exact complement of that ROI) on identical attention-eligible slices. Analyses involving cross-class matching were treated as secondary because the matching procedure necessarily used the released outcome labels to form balanced Sick/Normal pairs before classifier cross-validation. Manual-subset analyses were treated as mask-validation analyses rather than independent estimates of clinical performance.

2.2 Data source, permission, and release structure

The source was the CAD Cardiac MRI Dataset created by Danial Sharifrazi and distributed through Kaggle [2]. The associated article describes 63,648 images from a substantially larger cohort, collected at Tehran Omid Hospital with institutional ethics approval and written informed consent [1]. The downloaded release used here contained 63,425 JPEG files: 37,564 in Normal folders and 25,861 in Sick folders, arranged under 30 unique top-level Directory_* folders (16 Normal and 14 Sick). Immediate and nested folders were treated as 2,859 series proxies; because original DICOM metadata were absent, these proxies were not represented as verified SeriesInstanceUIDs.

The Kaggle license field was not sufficiently clear at project initiation. The author therefore contacted the dataset creator rather than assuming redistribution rights. Danial Sharifrazi granted academic-research use with citation of the associated article on 14 November 2025 and later confirmed the Directory_*-to-patient mapping and subset status on 2 September 2026 (personal correspondence). No source images are reproduced in this manuscript.

2.3 Structural audit and leakage control

Patient identity was defined as the top-level Directory_* folder. All descendant images and series proxies from one patient remained together throughout segmentation and classification splitting. A preliminary audit of the same release identified 1,094 exact duplicate groups but no exact duplicate group spanning patients or class labels. A perceptual-hash screen generated 24 cross-patient candidate edges, 10 of which crossed class labels; these were retained as screening candidates rather than treated as confirmed duplicates. This unresolved near-duplicate screen is reported as a limitation. A fixed seed of 42 was used for deterministic fold construction and model-selection procedures.

2.4 Image preprocessing and quality assessment

Images were read in grayscale. Near-black, low-variance scanner padding was removed conservatively while retaining at least 60% of each original dimension and limiting removal to 20% per side. Intensities were clipped to the 1st and 99th percentiles, scaled to [0,1], resized with preserved aspect ratio, and centred on a 256x256 canvas. For feature extraction, views were resized to 224x224.

Sharpness, noise ratio, dynamic range, and a perceptual hash were calculated for every image. Quality thresholds combined fixed safeguards with patient-specific tail quantiles. The audit covered all 63,425 images and was persisted under a stable dataset fingerprint. Downstream target training, attention-eligible cohorts, and matching required the relevant quality-valid flag; no CAD label was used to define the image-quality measures.

2.5 Manual targets and review workflow

A notebook-based HTML editor supported three semantic target states: HEART_PRESENT, NO_HEART_VISIBLE, and UNUSABLE. The current manual audit retained 5,097 training targets across all 30 patients and all five folds: 3,949 heart-present masks and 1,148 no-heart targets. A further 161 targets were rejected because of empty or unlabelled masks or image-quality failures, and 1,121 were explicitly marked unusable and excluded. HEART_PRESENT required a non-trivial binary mask; NO_HEART_VISIBLE required an empty target. Default target weights were 1.00 for manually accepted positive masks, 0.90 for no-heart targets, and 0.72 for confirmed-auto sources where applicable.

2.6 Patient-level cross-fitted 2.5D Attention U-Net

The segmentation network was an Attention U-Net [3,4] with three input channels representing the previous, current, and next image within the same folder-defined sequence. A second output head predicted whether the heart was visible. Training combined focal loss, Tversky loss (with stronger penalty on false-positive mask area), and an auxiliary heart-presence loss. Patient- and series-aware weighted sampling reduced domination by scans with many similar frames. Data augmentation included small rotations, scale and translation changes, intensity perturbations, blur, and noise.

Five segmentation folds were created at patient level. For each target fold, model fitting and threshold calibration excluded every patient in that fold. A separate validation-patient subset was selected only from non-test patients. The held-out fold was then predicted once, yielding a genuinely out-of-fold mask for every downstream image. Mask plausibility checks evaluated area, peak probability, boundary contact, and deviation from geometry priors derived from non-test masks. Invalid initial masks were subjected to threshold repair; uncertainty combined entropy, photometric TTA disagreement, presence ambiguity, geometry deviation, and sequence inconsistency.

2.7 Same-slice controls, cross-class matching, and manual-reference cohorts

Attention-eligible images had valid quality, a valid final out-of-fold mask, predicted heart presence, and mask area of at least 0.3% of the image. Exactly the same 50,735 slices and 2,541 series proxies were used for B2 (full image), AU1 (automatic ROI), and C1 (its complement). Before feature extraction, the automatic mask was expanded with a 15 x 15 max-pooling support; the complement retained exactly the visible image content outside that support.

For the secondary matched analysis, quality-valid and attention-valid Sick and Normal images were grouped into shared acquisition families using descriptors derived from perceptual hash, mask geometry, within-series position, and image quality. The current matching run selected 8,875 one-to-one pairs: 3,267 strict-core pairs plus 5,608 capacity-controlled extended pairs, yielding 17,750 images and 2,173 series proxies across all 30 patients. Patient, sequence, and anatomical caps prevented the requested target from being filled completely. Because class labels were required to construct Sick-Normal pairs, these modes are secondary sensitivity analyses rather than independent prospective estimates.

The manual-positive cohort contained the same 3,949 images and 1,257 series proxies for four representations: the accepted manual ROI (M1), its complement (C3), the automatic ROI on those images (AU3), and its complement (C4). This alignment supported paired automatic-versus-manual and complement-versus-complement comparisons without changing the source images.

2.8 Frozen EfficientNet representations and patient aggregation

An ImageNet1K V1-pretrained EfficientNet-B0 was used as a frozen 1,280-dimensional embedding extractor [5,6]; no EfficientNet parameter was updated using CAD labels. To match the defined experiment protocol, extraction ran on CPU in float32 with autocast disabled, batch size 4, and forward-batch size 4. Eleven representations were computed: one all-slice baseline, three attention-eligible same-slice views, three matched views, and four manual-positive validation views. Slice embeddings were averaged within each folder-defined series, and the series means were then averaged to form one vector per patient so that long series did not dominate solely through slice count.

2.9 Patient-level classifier and statistical analysis

All reported classification outputs were out-of-fold at patient level. Five stratified outer folds were shared across representations. Within each outer training set, an inner stratified cross-validation selected the logistic-regression regularization value C from $\{0.01, 0.1, 1, 10\}$ and selected a probability threshold by the Youden index [14]. The final pipeline applied standardization, PCA retaining 95% of variance, and class-balanced logistic regression implemented in scikit-learn [8].

Performance measures were ROC AUC, 95% patient-bootstrap AUC interval, average precision, Brier score, accuracy, sensitivity, and specificity. Confidence intervals used 2,000 bootstrap resamples of the 30 patient-level out-of-fold scores [15]. Pairwise representation comparisons merged scores by patient and class and bootstrapped the paired AUC difference with 2,000 resamples. No multiplicity adjustment was applied; pairwise analyses are exploratory, and confidence intervals rather than dichotomous significance labels are emphasized.

2.10 Implementation and reproducibility

The pipeline was implemented in Python with PyTorch/torchvision [7], OpenCV, NumPy, pandas, and scikit-learn [8] in Kaggle. All numerical values in this manuscript are transcribed from the complete result set archived on 5 October 2026 in the supplied notebook and console transcripts. The notebook main code cell and the supplied Python source are byte-identical. The expanded target state invalidated all five Attention checkpoints; all five out-of-fold prediction parts were recomputed. The final CPU stage then detected a changed semantic feature fingerprint, rebuilt the complete eleven-mode CPU/float32 feature bank, and

regenerated the patient-level evaluation. Source, output, and manuscript checksums are included in the submission package.

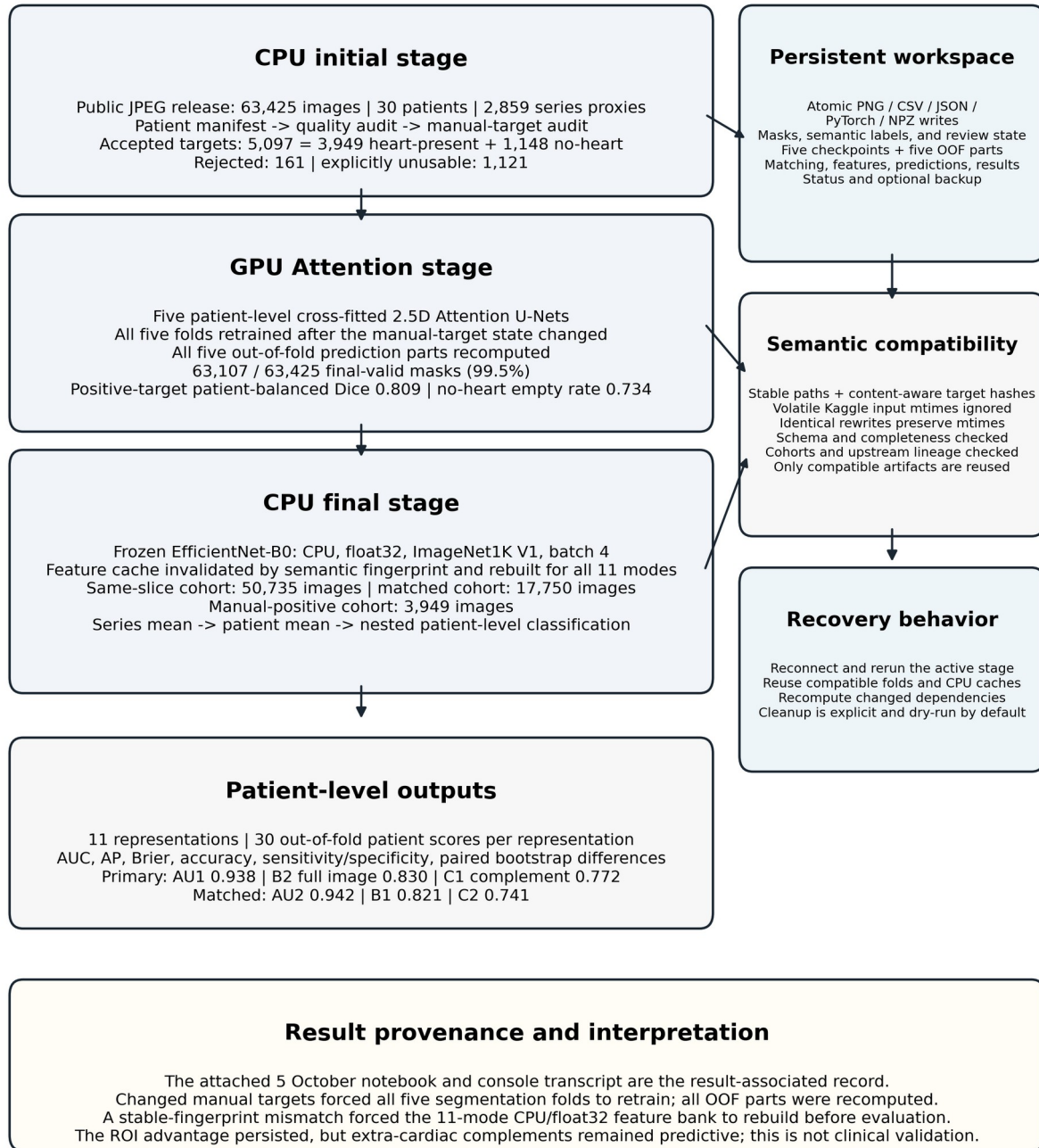
2.11 Persistent execution architecture and recovery

The current implementation separates Kaggle execution into three restart-safe sessions: (i) an initial CPU stage for dataset discovery, the dataset manifest, image-quality audit, and manual-target audit; (ii) a GPU stage for five cross-fitted Attention U-Net checkpoints and per-fold out-of-fold predictions; and (iii) a final CPU stage for Sick-Normal matching, the eleven-mode EfficientNet feature bank, patient-level classification, and optional HTML review. Each stage can be rerun after a disconnection without automatically deleting its persistent workspace.

PNG, CSV, JSON, PyTorch, and NPZ artifacts are written atomically. Identical PNG, CSV, and JSON content is left untouched so that harmless rewrites do not change modification times. Dataset identities use mount-independent paths beginning at `Directory_*`, patient and series metadata, neighbour context, and file size while deliberately ignoring volatile Kaggle-input mtimes. Manual-state fingerprints hash quality state, semantic label, source, weight, and mask bytes. Checkpoints and per-fold prediction parts require compatible schemas, complete expected tokens and masks, and matching stable fingerprints; legacy artifacts can be adopted only after additional split, count, hash, and freshness checks. Matching, feature, and evaluation caches validate semantic upstream lineage and cohort structure. In the reported run, these checks correctly forced all training and prediction folds, and then the feature bank, to be recomputed after the annotations changed.

Leakage-controlled analysis with semantic restart safety

Counts and performance are from the complete 5 October 2026 execution.



OOF = out-of-fold; ROI = region of interest; AP = average precision.

Figure 1. Leakage-controlled analysis workflow and semantic restart-safe execution architecture for the complete 5 October 2026 run. B = full-image baseline; AU = automatic Attention U-Net ROI; C = complement/control; M = accepted manual ROI. The figure distinguishes scientific stages, persistent artifacts, compatibility rules, and the concrete invalidation events that caused all five segmentation folds and the eleven-mode feature bank to be rebuilt. The matched analysis is secondary because class labels were used to construct pairs.

3. Results

3.1 Dataset, quality audit, and manual targets

The released dataset comprised 63,425 images from 30 patients (16 Normal, 14 Sick) and 2,859 folder-defined series proxies. The complete image-quality audit was reused under a compatible stable dataset fingerprint. The expanded manual-target audit retained 5,097 training targets: 3,949 heart-present masks and 1,148 no-heart targets. It rejected 161 targets and excluded 1,121 explicitly unusable targets. Accepted targets covered all 30 patients and all five segmentation folds. Relative to the earlier manuscript version, the target set was substantially expanded; because all dependent models, predictions, cohorts, features, and classifiers were regenerated, version-to-version changes cannot be attributed to the additional masks alone.

Stage or quantity	Value	Denominator	Interpretation
Released patient units	30	-	16 Normal; 14 Sick
Released JPEG images	63,425	-	Effective n is patients, not images
Folder-defined series proxies	2,859	-	Not verified DICOM SeriesInstanceUIDs
Accepted segmentation targets	5,097	-	3,949 heart-present; 1,148 no-heart
Accepted heart-present targets	3,949	5,097	Usable positive masks across all patients
Accepted no-heart targets	1,148	5,097	Explicit empty segmentation targets
Rejected / explicitly unusable	161 / 1,121	-	Rejected by mask/quality rules / excluded by label
Attention-eligible slices	50,735	63,425	Shared exactly by B2, AU1, and C1

Table 1. Cohort and target-quality summary for the complete 5 October 2026 run.

3.2 Out-of-fold segmentation performance

The changed manual-target state invalidated every prior fold checkpoint, so all five 2.5D Attention U-Nets were retrained. All five out-of-fold prediction parts were then recomputed rather than reused. Across 63,425 released images, 63,107 final masks passed the validity rules and 318 remained invalid. Against 5,097 accepted targets, the patient-balanced Dice score on heart-present targets was 0.8088. Among accepted no-heart targets, 73.4% received an appropriately empty prediction; thus approximately one quarter still produced non-empty false-positive masks. These values support useful localization while documenting a persistent failure mode on frames without a visible heart.

Metric	Value	Level / subset
Evaluated accepted targets	5,097	OOF manual-target evaluation
Positive-target patient-balanced Dice	0.8088	Heart-present targets
Empty-prediction rate	0.7343	Accepted no-heart targets
Final-valid automatic masks	63,107 / 63,425	All released images
Remaining invalid masks	318 / 63,425	All released images
Segmentation folds retrained	5 / 5	Changed target state
OOF prediction folds recomputed	5 / 5	No fold part reused

Table 2. Out-of-fold Attention U-Net performance and recomputation summary for the current run.

3.3 Analysis cohorts

The all-slice baseline contained 63,425 images and 2,859 series proxies. The primary same-slice attention-eligible cohort contained 50,735 images and 2,541 series proxies. The secondary matched cohort contained 17,750 images (8,875 Sick-Normal pairs) and 2,173 series proxies. The manual-positive validation cohort contained 3,949 images and 1,257 series proxies. Every cohort included all 30 patients, although selected slice and series distributions differed by cohort.

Cohort	Modes	Patients	Slices	Series proxies
All released slices	B0	30	63,425	2,859
Attention-eligible same slices	B2, AU1, C1	30	50,735	2,541
Cross-class matched	B1, AU2, C2	30	17,750	2,173
Manual-positive same images	M1, C3, AU3, C4	30	3,949	1,257

Table 3. Source cohorts used to construct the eleven current patient representations.

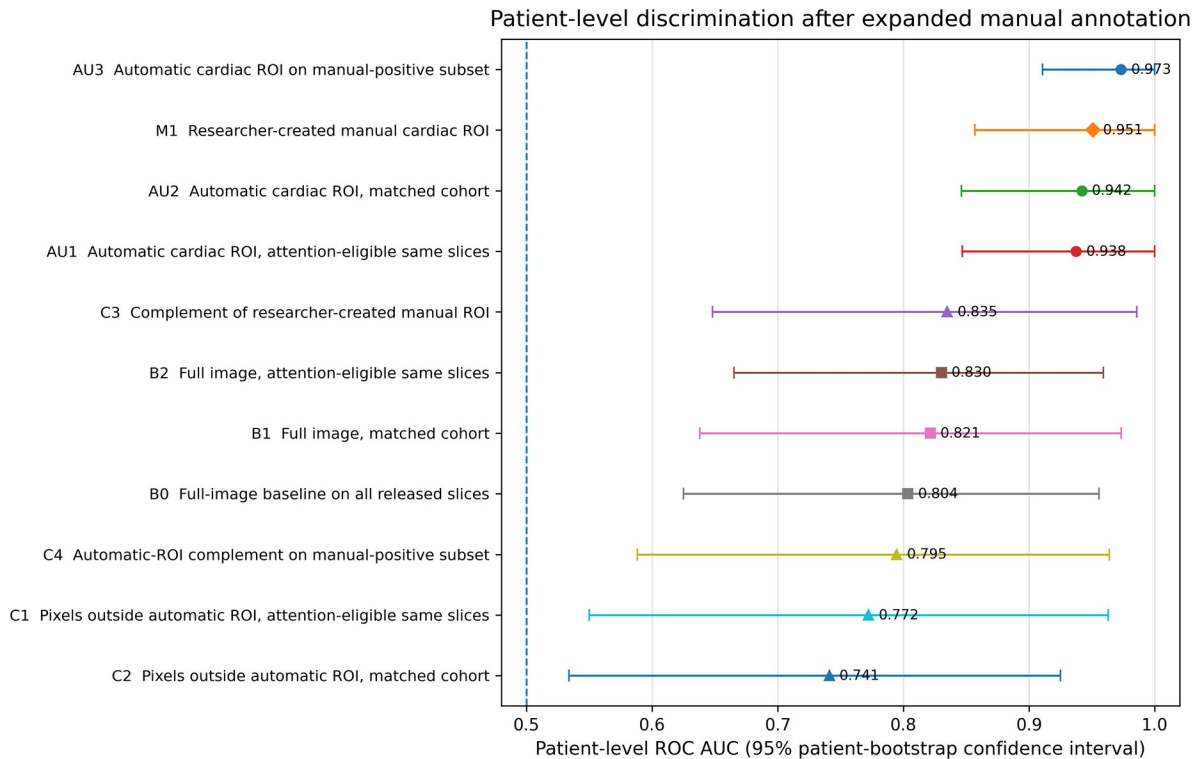
3.4 Patient-level classification

On the primary attention-eligible cohort, the automatic ROI (AU1) achieved AUC 0.938 (95% CI 0.847-1.000), AP 0.938, and Brier score 0.104. On the exact same slices, the full-image control (B2) achieved AUC 0.830 and the complement control (C1) AUC 0.772. The all-slice baseline (B0) achieved AUC 0.804. In the secondary matched cohort, AU2 achieved AUC 0.942, compared with 0.821 for B1 and 0.741 for C2. On the manual-positive cohort, automatic ROI AU3 achieved AUC 0.973 and manual ROI M1 achieved AUC 0.951. The complement of the manual ROI (C3) remained predictive at AUC 0.835.

Mode	Representation	Slices	AUC	95% CI	AP	Brier	Accuracy
AU3	Automatic cardiac ROI on manual-positive subset	3,949	0.973	0.911-1.000	0.975	0.069	0.900
M1	Accepted manual cardiac ROI	3,949	0.951	0.857-1.000	0.957	0.077	0.833
AU2	Automatic cardiac ROI, matched cohort	17,750	0.942	0.846-1.000	0.943	0.101	0.900
AU1	Automatic cardiac ROI, attention-eligible same slices	50,735	0.938	0.847-1.000	0.938	0.104	0.800
C3	Complement of accepted manual ROI	3,949	0.835	0.648-0.986	0.888	0.159	0.800
B2	Full image, attention-eligible same slices	50,735	0.830	0.665-0.959	0.829	0.167	0.733
B1	Full image, matched cohort	17,750	0.821	0.638-0.974	0.864	0.158	0.833
B0	Full-image baseline on all released slices	63,425	0.804	0.625-0.956	0.817	0.165	0.833

Mode	Representation	Slices	AUC	95% CI	AP	Brier	Accuracy
C4	Automatic-ROI complement on manual-positive subset	3,949	0.795	0.588-0.964	0.837	0.175	0.800
C1	Pixels outside automatic ROI, attention-eligible same slices	50,735	0.772	0.550-0.963	0.855	0.173	0.733
C2	Pixels outside automatic ROI, matched cohort	17,750	0.741	0.534-0.925	0.785	0.234	0.733

Table 4. Patient-level out-of-fold classification results from the complete 5 October 2026 run. AUC intervals use 2,000 patient-level bootstrap resamples. Sensitivity and specificity are shown. The machine-readable supplement reproduces the aggregate evaluation and paired-comparison tables; decision thresholds and individual patient predictions remain in the persisted pipeline workspace and were not available in the supplied export, so they are not redistributed with this preprint package. Matched and manual-subset modes are secondary or validation analyses.



Complete 5 October 2026 run; n = 30 patients; 2,000 patient-level bootstrap resamples.
Circles = automatic ROI; diamond = manual ROI; squares = full image;
triangles = complement/control.

Figure 2. Patient-level ROC AUCs with 95% patient-bootstrap confidence intervals after expanded manual annotation and complete recomputation. ROI representations rank above their aligned full-image and complement controls, but intervals remain wide because the independent sample comprises 30 patients. The plot uses the final feature-bank cohorts shown in Table 3.

3.5 Paired same-patient comparisons

Within the primary attention-eligible cohort, AU1 exceeded B2 by 0.107 AUC (95% paired bootstrap CI 0.009-0.235) and exceeded C1 by 0.165 (0.029-0.344). B2 exceeded C1 by 0.058 (-0.110 to 0.244), an interval compatible with little or no difference between the full image and extra-cardiac complement. In the matched cohort, AU2 exceeded B1 by 0.121 (0.010-0.262) and C2 by 0.201 (0.054-0.384).

On the manual-positive subset, the AUC difference between manual and automatic ROIs was -0.022 (M1 minus AU3; 95% CI -0.081 to 0.009). The interval includes zero and does not establish formal equivalence, although it excludes a large advantage for the manual ROI in this sample. Complement definitions were also not detectably different (C3 minus C4 = 0.040; 95% CI -0.081 to 0.176).

Comparison	First - second	Patients	AUC difference	95% CI
Automatic ROI vs same-slice full image	AU1 - B2	30	0.107	0.009 to 0.235
Automatic ROI vs same-slice complement	AU1 - C1	30	0.165	0.029 to 0.344
Matched automatic ROI vs matched full image	AU2 - B1	30	0.121	0.010 to 0.262
Matched automatic ROI vs matched complement	AU2 - C2	30	0.201	0.054 to 0.384
Manual ROI vs automatic ROI on same images	M1 - AU3	30	-0.022	-0.081 to 0.009
Manual complement vs automatic complement	C3 - C4	30	0.040	-0.081 to 0.176

Table 5. Key paired AUC comparisons from the current run. Differences are exploratory and unadjusted for multiplicity; the complete paired comparison set is shown in Supplementary Table S2.

3.6 Residual shortcut signal

The central warning signal remained the predictive performance of complement controls. C1 achieved AUC 0.772 on the attention-eligible same-slice cohort, C2 achieved 0.741 after cross-class matching, C3 achieved 0.835 outside the accepted manual ROI, and C4 achieved 0.795 outside the automatic ROI on those same annotated images. The full-image-minus-complement differences did not exclude zero in either the primary or matched cohort. Matching changed complement AUC by only 0.031 (C1 minus C2; 95% CI -0.079 to 0.142). These findings remain consistent with label-associated acquisition style, sequence distribution, support geometry, surrounding anatomy, JPEG export, or other release-specific context. The study cannot identify the dominant mechanism because original DICOM metadata and patient covariates were unavailable.

4. Discussion

4.1 Principal findings

This study contributes a leakage-controlled reanalysis of a public cardiac MRI release whose image count greatly exceeded its number of independent patients. After the manual annotation set was expanded and every affected computational stage was rerun, the primary result remained an advantage for the out-of-fold cardiac ROI over both the full image and the exact extra-cardiac complement on the same source slices. The paired intervals for AU1 versus B2 and AU1 versus C1 excluded zero in the current run, supporting a within-release benefit from cardiac localization under the implemented protocol.

At the same time, complement representations remained predictive, with AUCs from 0.741 to 0.835. This prevents the interpretation that the classifier learned only cardiac pathology. In a dataset with 30 patients, stable patient, scanner, sequence, export, or mask-support patterns can correlate with the folder label. ROI restriction reduced this signal in the primary and matched comparisons but did not eliminate it. The defensible conclusion is therefore dual: cardiac localization appears useful within this release, and important non-cardiac shortcut information remains.

4.2 Why patient-level and out-of-fold design matters

Treating 63,425 frames as independent examples would obscure the effective sample size and permit neighbouring images from one patient to appear on both sides of a split. The pipeline instead kept each patient intact, generated segmentation masks without exposing the held-out patient to the segmentation model, and evaluated classification only after series- and patient-level pooling. These safeguards address common leakage mechanisms in machine-learning science [9] and make the reported AUCs substantially more interpretable than image-level accuracy alone. They do not, however, solve sampling bias, missing covariates, or single-release overfitting.

4.3 Automatic versus researcher-created regions

Automatic ROI AUC on the manual-positive subset was numerically higher than manual ROI AUC, while the paired interval included zero. This suggests that automatic masks were not detectably inferior for classification in the present sample. It does not prove anatomical superiority: the classifier may benefit from systematic support shape, dilation, or context retained by the automatic ROI. Manual targets were produced or reviewed by a single non-radiologist researcher and were not independently adjudicated. Positive-target patient-balanced Dice of 0.809 and a 73.4% no-heart empty-prediction rate demonstrate useful localization together with remaining false-positive behavior. The current target expansion triggered a full segmentation and classification rerun, so differences from prior versions are descriptive rather than a controlled estimate of annotation-set size.

4.4 Interpretation of matched-cohort analyses

The matched analyses attempted to control acquisition and anatomical differences by pairing Sick and Normal images within shared descriptor families. The ROI retained an advantage over full and complement representations after this procedure. However, matching used class labels to choose pairs before outer classifier evaluation. Consequently, AU2, B1, and C2 should not be interpreted as independent prospective performance estimates; they are secondary sensitivity analyses that ask whether the ROI advantage persists after an outcome-informed balancing procedure. A future version should reconstruct matching strictly within each outer training fold and apply the learned matching rule to held-out patients without consulting their labels.

4.5 Strengths

- The unit of analysis was corrected from images to confirmed patient folders.
- Segmentation was patient-level cross-fitted, so every downstream mask was out-of-fold for its patient.
- Primary ROI, full-image, and complement comparisons used exactly the same slices and patients.
- Patient features were balanced first across series and then across patients.
- Paired patient-level bootstrap intervals quantified uncertainty in representation differences.
- Negative controls were interpreted as evidence about shortcut risk rather than ignored when they remained predictive.
- The result-associated software preserves stage provenance through semantic, content-aware fingerprints, atomic idempotent writes, structural cohort checks, and per-fold recovery. The current execution demonstrated the intended behavior: changed targets invalidated all five checkpoints and prediction parts, while a changed upstream lineage invalidated and rebuilt the eleven-mode feature bank before evaluation.

4.6 Limitations

- Only 30 independent patients were available. Confidence intervals are wide and may be unstable, and no external test cohort was evaluated.
- The release contained JPEG exports rather than original DICOM files; sequence names, scanner parameters, demographics, comorbidities, and site metadata were unavailable for adjustment or subgroup analysis.

- Folder labels were used as the outcome. The study did not independently verify angiographic status or clinical diagnostic criteria for the 30-patient subset.
- Research masks were produced by one author and were not radiologist-adjudicated; formal annotation blinding was not recorded.
- Twenty-four perceptual cross-patient candidate edges remained unconfirmed, although no exact cross-patient duplicates were found.
- The secondary matched cohort was selected using class labels before outer evaluation and therefore may be outcome-informed.
- Hyperparameter choices, quality gates, mask-repair rules, and representation definitions were developed on the same release, creating researcher-degrees-of-freedom and release-specific overfitting risk.
- No multiplicity correction was applied across paired comparisons, and all inferential results are exploratory.
- The result package preserves source code, notebook outputs, console transcripts, parsed aggregate tables, and SHA-256 hashes, but it does not contain a complete frozen Kaggle environment or container image. CUDA deterministic algorithms were not enforced and cuDNN benchmarking was enabled, so bit-identical replay is not guaranteed across hardware, drivers, or library versions even when semantic cache checks pass.

4.7 Future validation

The next stage should use an independently governed clinical DICOM cohort with patient-level labels, radiologist-reviewed cardiac masks, complete acquisition metadata, and a locked analysis plan. Sequence-aware models should distinguish cine, perfusion, late-gadolinium enhancement, T2-weighted, and localizer images rather than relying on folder proxies. Matching or harmonization must be learned inside each outer training fold and applied without consulting held-out labels. External evaluation should report calibration, decision-curve utility, subgroup performance, failure analysis, and uncertainty, and should include negative controls that directly test scanner, padding, support-shape, and extra-cardiac shortcuts. Future updates should archive a complete software environment in addition to the current source, artifacts, and checksums. Until independent validation is complete, the present model should be regarded solely as a methodological proof of concept.

5. Conclusion

In a 30-patient public cardiac MRI release, out-of-fold automatic cardiac ROIs improved patient-level CAD-label discrimination relative to exact same-slice full-image and complement controls after expanded manual annotation and complete recomputation. The advantage persisted in an exploratory matched analysis, and automatic ROI performance was not detectably different from manual ROI performance on the same annotated images. Nevertheless, complement AUCs of 0.741-0.835, the very small number of independent patients, unavailable DICOM metadata, single-annotator targets, outcome-informed matching, and absent external validation demonstrate substantial residual uncertainty. The appropriate conclusion is not clinical readiness, but evidence that cardiac localization is promising while dataset-specific shortcuts remain material and require independent validation.

Declarations

Ethics approval and consent to participate

The original data collection was reported to have received approval from the Tehran Omid Hospital ethics committee, and all participants provided written informed consent [1]. The present study was a secondary computational analysis of a publicly accessible image release without participant contact or access to direct identifiers. No additional human-participant data were collected, and no additional institutional ethics approval was obtained for this reanalysis.

Consent for publication

Not applicable. No identifiable participant information or source image is reproduced in this manuscript.

Data availability

The analysed images are available from the CAD Cardiac MRI Dataset on Kaggle: [CAD Cardiac MRI Dataset repository](#). The dataset creator granted the author academic-research use with citation by personal correspondence. This manuscript does not redistribute source images.

Code availability

The result-associated implementation and its supporting documentation are now publicly available through the executable Kaggle notebook and the version-controlled GitHub repository listed below. The supplementary package supplies the result-associated Python source and executed Jupyter notebook dated 5 October 2026. The notebook main code cell is byte-identical to the Python file. The package also includes the final CPU console transcript, an extracted GPU training and out-of-fold prediction transcript, machine-readable evaluation and paired-comparison tables, run and segmentation summaries, figures, a provenance note, a manifest, and SHA-256 checksums. The current source implements separate initial-CPU, GPU, and final-CPU entry points; atomic idempotent writes; semantic stable fingerprints; safe legacy-cache adoption; per-fold recovery; status, backup, and explicit dry-run cleanup utilities; and CPU/float32 EfficientNet extraction. Source images, manual masks, trained checkpoints, and automatic mask PNGs are not redistributed in the preprint package.

Public project resources:

Executable Kaggle notebook: [Cardiac MRI CAD - leakage-controlled, resumable research pipeline](#)

GitHub source repository: [DavidMarghidan/cad-cardiac-mri-project-dvm](#)

GitHub documentation: [repository /docs directory and document index](#)

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Competing interests

The author declares no competing interests.

Author contributions

David Vlad Marghidan: conceptualization; data curation; formal analysis; investigation; methodology; software; validation; visualization; writing - original draft; writing - review and editing.

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Declaration of generative-AI assistance

OpenAI ChatGPT was used as an assistive tool during code-review and refactoring discussions, debugging, documentation, language editing, manuscript organization, comparison of source versions, extraction and consistency checking of aggregate author-supplied run outputs, and preparation of explanatory figure

layouts. For this version, AI assistance was used to synchronize the manuscript with the supplied 5 October notebook and console transcripts and to flag values not supported by the current run record. The AI system did not execute the Kaggle training run and did not generate, estimate, or impute the reported scientific metrics. Study design, dataset audit, annotations, software execution, validation decisions, scientific interpretation, reference verification, source-to-manuscript reconciliation, and final editorial responsibility remained with David Vlad Marghidan under human scientific supervision. No confidential clinical data, source patient images, or directly identifiable participant information were supplied to the tool. The AI system is not an author and cannot assume responsibility for the manuscript.

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Supplementary information

Supplementary Note S1. Experiment definitions

Mode names are fixed research labels. B denotes a full-image baseline, AU an automatic Attention U-Net ROI, C a complement/control region, and M a researcher-created manual ROI. All representations are pooled at series and patient level before classification.

Mode	Cohort	Representation	Role
B0	All slices	Robust-normalized full image	Broad reference
B2	Attention-eligible same slices	Full image	Primary same-slice control
AU1	Attention-eligible same slices	Automatic dilated cardiac ROI	Primary ROI
C1	Attention-eligible same slices	Exact visible-content complement of AU1	Primary negative control
B1	Matched	Full image	Secondary matched control
AU2	Matched	Automatic dilated cardiac ROI	Secondary matched ROI
C2	Matched	Automatic-ROI complement	Secondary matched negative control
M1	Manual-positive	Accepted manual dilated cardiac ROI	Mask reference
C3	Manual-positive	Manual-ROI complement	Mask-reference control
AU3	Manual-positive	Automatic ROI on the same images as M1	Automatic/manual comparison
C4	Manual-positive	Automatic complement on the same images as C3	Complement comparison

Supplementary Table S1. Definitions and inferential roles of the eleven experiment modes.

Supplementary Note S2. Full paired AUC comparison set

Comparison	First - second	n	AUC difference	95% CI
Automatic ROI vs same-slice full image	AU1 - B2	30	0.107	0.009 to 0.235
Same-slice full image vs complement	B2 - C1	30	0.058	-0.110 to 0.244
Automatic ROI vs same-slice complement	AU1 - C1	30	0.165	0.029 to 0.344
Attention-eligible vs all-slice full image	B2 - B0	30	0.027	-0.125 to 0.181
Matched automatic ROI vs matched full image	AU2 - B1	30	0.121	0.010 to 0.262
Matched automatic ROI vs matched complement	AU2 - C2	30	0.201	0.054 to 0.384
Matched full image vs matched complement	B1 - C2	30	0.080	-0.009 to 0.190
All-slice vs matched full image	B0 - B1	30	-0.018	-0.120 to 0.077
Attention-eligible vs matched automatic ROI	AU1 - AU2	30	-0.004	-0.060 to 0.040
Attention-eligible vs matched complement	C1 - C2	30	0.031	-0.079 to 0.142

Comparison	First - second	n	AUC difference	95% CI
Manual ROI vs automatic ROI on same images	M1 - AU3	30	-0.022	-0.081 to 0.009
Manual complement vs automatic complement	C3 - C4	30	0.040	-0.081 to 0.176

Supplementary Table S2. All paired patient-level AUC differences from the complete 5 October 2026 run. Confidence intervals use 2,000 paired patient-bootstrap resamples. No multiplicity correction was applied.

Supplementary Note S3. Historical structural audit of the same release

Before the current streamlined run, the same downloaded release underwent a broader duplicate audit to identify visual content that could cross patient boundaries even when folder identities did not. Exact decoded-pixel matches were treated as deterministic duplicate evidence; perceptual-hash matches were retained only as screening candidates.

Audit item	Result	Interpretation
Exact duplicate groups	1,094	All confined within patient and class
Exact cross-patient groups	0	No deterministic cross-patient duplicate detected
Exact cross-label groups	0	No deterministic cross-label duplicate detected
Unique perceptual hashes searched	30,225	Complete BK-tree radius search
Cross-patient perceptual candidates	24	Unconfirmed screening candidates
Cross-label perceptual candidates	10	Unconfirmed screening candidates

Supplementary Table S3. Historical duplicate-screening results for the same 63,425-image release. Perceptual candidates were not equivalent to confirmed duplicates.

Supplementary Note S4. Result and software provenance

All reported metrics were generated by the complete execution archived on 5 October 2026 using the supplied source and notebook. The accepted target state comprised 5,097 targets, including 3,949 heart-present masks. Because positive target counts changed, all five segmentation checkpoints were retrained; all five out-of-fold prediction parts were recomputed. The final CPU stage then rejected the previous feature bank under the stable semantic fingerprint and rebuilt all eleven representations before producing the current patient-level results. Earlier numerical values in manuscript version 1.1 are superseded and are not pooled with the current results. Run-to-run differences are descriptive because annotations, segmentation models, masks, eligibility, matching, feature representations, and classifier fits all changed together. Exact SHA-256 values for the source, notebook, transcripts, parsed tables, figures, and manuscripts are provided in SHA256SUMS.txt.

Supplementary Note S5. Restart-safe artifact contract

The current source defines an explicit semantic persistence contract. Dataset and manual-target identities are based on stable paths, content, labels, quality state, and mask hashes rather than volatile Kaggle mount times. Identical PNG, CSV, and JSON writes preserve existing files and mtimes. Checkpoints, prediction parts, matching, the feature bank, and evaluation outputs are reused only after schema, completeness, dependency, and cohort checks pass; otherwise the smallest affected stage is recomputed. The table below documents operational reproducibility and the behavior demonstrated by the reported run.

Stage / device	Persistent outputs	Semantic compatibility validation	Restart behavior
Initial CPU	Dataset manifest; quality CSV/JSON; manual audit CSV/JSON	Stable Directory_*-relative dataset identity; file size; schema and token completeness; content-aware manual label and mask hash	Reuse complete compatible audits; rebuild only the invalid audit
GPU training	Five Attention U-Net checkpoints; training summary	Model schema; stable mask/label/source/weight fingerprint; patient split and target counts; checkpoint hash; safe legacy freshness checks	Reuse compatible folds; retrain changed or unfinished folds
GPU OOF prediction	Five fold CSV/JSON parts; automatic mask PNGs; combined audit	Checkpoint SHA-256; stable v6 settings and image identities; expected tokens; metadata schema; complete mask set	Reuse compatible fold parts; recompute incomplete or changed folds
Final CPU	Matching manifest; feature NPZ/JSON; evaluation CSVs and patient predictions	Semantic upstream lineage; expected patient labels and cohort counts; finite matrices; feature/evaluation fingerprints	Resume matching, feature extraction, or evaluation; current run rebuilt all eleven modes
Review / operations	Manual labels; review history; overlays; stage status; optional ZIP backup	Explicit semantic targets; workspace contract; idempotent writes; cleanup preview and path-safety checks	Reopen queues after restart; cleanup remains explicit and dry-run by default

Supplementary Table S4. Persistent artifacts, semantic validation rules, and recovery behavior in the result-associated 5 October 2026 implementation.

Supplementary Note S6. Research Square submission declarations included

- Structured abstract, title, keywords, article type, author and primary institution information.
- Ethics approval and consent statement for the original data and the secondary analysis.
- Consent for publication statement.
- Data and code availability statements.
- Funding, competing-interests, author-contribution, acknowledgement, and generative-AI declarations.
- All figures and tables cited in the manuscript, with embedded high-resolution figure files.
- The result-associated source and executed notebook, GPU and final-CPU transcripts, parsed result tables, current figures, provenance note, revision changelog, package manifest, and SHA-256 checksum file.

END OF MANUSCRIPT